Module 3: PINN Formulation for the Alfvén-Wave Benchmark

MHD_PINN_proj1 Physics Notes

Revised to derive the reduced benchmark PINN step by step

These notes explain the first executable physics-informed neural network (PINN) stage in the project. The goal is not to pretend that the network already solves fully general two-dimensional ideal MHD. The goal is to build the first PINN honestly from a reduced subsystem that is still physically meaningful, analytically understood, and trainable. The benchmark used for that purpose is the same small-amplitude Alfvén-wave problem introduced in Module 2.

1 Symbol and variable table

Symbol	Name	Meaning
(x, y, t)	coordinates and time	Spatial coordinates and time presented to the network.
ρ	mass density	Plasma density. In the reduced benchmark it remains at ρ_0.
u_x, u_y, u_z	velocity components	Components of the plasma bulk velocity. Only u_z is dynamically active in the reduced benchmark.
p	gas pressure	Thermal pressure. In the reduced benchmark it remains at p_0.
B_x, B_y, B_z	magnetic-field components	Magnetic-field components. Only B_z is dynamically active in the reduced benchmark; $B_x = B_0$ remains constant.
ρ_0, p_0, B_0	background state values	Uniform equilibrium values used by the Alfvén benchmark.
v_A	Alfvén speed	Propagation speed $v_A = B_0/\sqrt{\rho_0}$ in the normalized system.
$\mathcal{N}_\theta$	neural network	Parametric map from coordinates to predicted fields, with trainable parameters θ.
$\mathcal{L}_{\mathrm{IC}}$	initial-condition loss	Penalizes mismatch with the known state at $t = 0$.
$\mathcal{L}_{\mathrm{BC}}$	boundary-condition loss	Penalizes mismatch between periodic boundaries.
$\mathcal{L}_{\mathrm{PDE}}$	PDE-residual loss	Penalizes failure to satisfy the reduced Alfvén dynamics.
$\mathcal{L}_y$	y-uniformity loss	Penalizes dependence on y in a benchmark that should be one-dimensional.
$\mathcal{L}_{\mathrm{bg}}$	background-state loss	Penalizes drift of quantities that should stay constant.

Symbol	Name	Meaning
$\mathcal{L}_{\nabla\cdot B}$	divergence loss	Penalizes violation of $\nabla \cdot \mathbf{B} = 0$.

2 Why the first PINN should not begin from the full eight-equation system

A fully general two-dimensional ideal-MHD PINN is possible in principle, but it is not the best first scientific step. The reasons are physical, numerical, and methodological.

2.1 Physical reason

The full ideal-MHD system couples multiple wave families:

- Alfvén waves,
- slow magnetosonic waves,
- fast magnetosonic waves,
- entropy-like advective behavior,
- the magnetic-divergence constraint.

If a first PINN fails on that full system, it becomes hard to tell whether the problem is:

- the choice of network architecture,
- the loss design,
- poor optimization,
- bad collocation sampling,
- or a misunderstanding of the underlying benchmark physics.

2.2 Methodological reason

A project should first ask a narrow question that can be answered clearly:

Can a PINN reproduce the simplest physically nontrivial MHD benchmark if the relevant residual system is encoded correctly?

The Alfvén benchmark is ideal for that because its reduced dynamics can be derived explicitly from ideal MHD and its exact solution is known.

Design principle. The first PINN should be strong enough to be a real physics problem, but narrow enough that success or failure can be interpreted cleanly.

3 Reduced Alfvén dynamics derived from ideal MHD

3.1 Starting point: the benchmark background state

The equilibrium is

$$\rho = \rho_0, \qquad p = p_0, \qquad \mathbf{u}_0 = (0,0,0), \qquad \mathbf{B}_0 = (B_0, 0, 0). \tag{1}$$

The perturbation is carried by

$$u_z(x,t), \qquad B_z(x,t), \tag{2}$$

while the other fields remain at their background values.

3.2 Reduction assumptions

The benchmark is built so that the following statements are valid:

1. the solution is uniform in y,
2. the perturbation amplitude is small,
3. density and pressure remain constant to leading order,
4. $u_x = u_y = 0$ and $B_y = 0$,
5. only the transverse pair (u_z, B_z) evolves dynamically.

These assumptions reduce the full ideal-MHD system to the pair

$$\frac{\partial u_z}{\partial t} - \frac{B_0}{\rho_0}\frac{\partial B_z}{\partial x} = 0, \tag{3}$$

$$\frac{\partial B_z}{\partial t} - B_0 \frac{\partial u_z}{\partial x} = 0. \tag{4}$$

The derivation was already shown in Module 2; here the main point is that the PINN loss should be built from the *physics that actually governs this benchmark*, not from unrelated residuals.

3.3 Residual form

Move each equation to one side:

$$R_{u_z}(x,t) = \frac{\partial u_z}{\partial t} - \frac{B_0}{\rho_0}\frac{\partial B_z}{\partial x}, \tag{5}$$

$$R_{B_z}(x,t) = \frac{\partial B_z}{\partial t} - B_0 \frac{\partial u_z}{\partial x}. \tag{6}$$

The exact solution makes both residuals zero at every point.

4 What the network predicts and why

The network takes a point in space-time and returns eight fields:

$$\mathcal{N}_\theta(x,y,t) = (\rho, u_x, u_y, u_z, p, B_x, B_y, B_z). \tag{7}$$

At first that may seem wasteful. If only u_z and B_z are dynamically active, why predict the full state?

The answer is architectural continuity. The eventual goal of the project is a broader MHD PINN, so it is useful for the interface to already look like a full MHD predictor. The reduced benchmark is then enforced through the loss, not by hard-wiring the output dimension down to two variables.

5 How a PINN enforces physics

5.1 General idea

A PINN is different from ordinary supervised learning. In ordinary supervised learning, a network is shown many input-output pairs and is trained to reproduce them. In a PINN, the network is also penalized for violating differential equations and constraints. That means the loss function is not just a data mismatch; it is a statement of what the predicted fields are supposed to *be physically allowed to do*.

5.2 Collocation points

Choose many sample points in the domain:

$$\{(x^{(n)}, y^{(n)}, t^{(n)})\}_{n=1}^{N_c}. \tag{8}$$

At each point, evaluate the network output and its derivatives using automatic differentiation. Those derivatives are inserted into the residual formulas (**??**) and (**??**). This is how the network is forced to "feel" the PDE.

6 Loss terms in detail

6.1 Initial-condition loss

At $t = 0$, the benchmark state is known exactly:

$$u_z(x, 0) = A\sin(kx), \qquad B_z(x, 0) = -\sqrt{\rho_0}A\sin(kx), \tag{9}$$

with the remaining fields fixed at background values. Therefore

$$\begin{aligned}\mathcal{L}_{\mathrm{IC}} = {} & \mathrm{MSE}(\rho, \rho_0) + \mathrm{MSE}(u_x, 0) + \mathrm{MSE}(u_y, 0) + \mathrm{MSE}(p, p_0) \\ & + \mathrm{MSE}(B_x, B_0) + \mathrm{MSE}(B_y, 0) + \mathrm{MSE}(u_z, A\sin(kx)) + \mathrm{MSE}(B_z, -\sqrt{\rho_0}A\sin(kx)).\end{aligned} \tag{10}$$

This term anchors the network to the correct starting state.

6.2 Periodic-boundary loss

The conventional benchmark uses periodic boundaries in x. The PINN should mirror that physics:

$$q(x_{\min}, y, t) = q(x_{\max}, y, t) \tag{11}$$

for each predicted field q. So one natural loss is

$$\mathcal{L}_{\mathrm{BC}} = \sum_q \mathrm{MSE}(q(x_{\min}, y, t),\ q(x_{\max}, y, t))\,. \tag{12}$$

This is not prescribing the value at the boundary; it is enforcing the topology of the periodic domain.

6.3 Reduced-PDE loss

At collocation points inside the domain, the reduced Alfvén residuals should vanish. Therefore

$$\mathcal{L}_{\mathrm{PDE}} = \mathrm{MSE}(R_{u_z}, 0) + \mathrm{MSE}(R_{B_z}, 0). \tag{13}$$

This is the core physics term for the first executable PINN. It is where the benchmark dynamics enters.

6.4 y-uniformity loss

Although the network accepts (x, y, t) as input, the benchmark solution is independent of y. So the network should also learn

$$\frac{\partial q}{\partial y} = 0 \tag{14}$$

for every predicted field q. One way to enforce that is

$$\mathcal{L}_y = \sum_q \mathrm{MSE}\left(\frac{\partial q}{\partial y}, 0\right). \tag{15}$$

This is not a random regularizer. It is a physics statement about the benchmark symmetry.

6.5 Background-state loss

Several fields should stay fixed throughout the reduced benchmark:

$$\rho = \rho_0, \qquad p = p_0, \qquad B_x = B_0, \qquad u_x = u_y = B_y = 0. \tag{16}$$

Those conditions can be encouraged away from the initial slice through

$$\begin{aligned}\mathcal{L}_{\mathrm{bg}} = {} & \mathrm{MSE}(\rho, \rho_0) + \mathrm{MSE}(p, p_0) + \mathrm{MSE}(B_x, B_0) \\ & + \mathrm{MSE}(u_x, 0) + \mathrm{MSE}(u_y, 0) + \mathrm{MSE}(B_y, 0).\end{aligned} \tag{17}$$

This prevents the network from inventing spurious compressive or oblique structure that does not belong to the benchmark.

6.6 Divergence-free magnetic-field loss

Ideal MHD requires

$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = 0. \tag{18}$$

For the exact benchmark this is trivially true because $B_x = B_0$ and $B_y = 0$. Even so, it is worth keeping the term

$$\mathcal{L}_{\nabla \cdot B} = \mathrm{MSE}\left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y}, 0\right), \tag{19}$$

because divergence control is structurally central to all later MHD models.

6.7 Total loss

A weighted total loss may be written schematically as

$$\mathcal{L}_{\mathrm{total}} = w_{\mathrm{IC}}\mathcal{L}_{\mathrm{IC}} + w_{\mathrm{BC}}\mathcal{L}_{\mathrm{BC}} + w_{\mathrm{PDE}}\mathcal{L}_{\mathrm{PDE}} + w_y\mathcal{L}_y + w_{\mathrm{bg}}\mathcal{L}_{\mathrm{bg}} + w_{\nabla \cdot B}\mathcal{L}_{\nabla \cdot B}. \tag{20}$$

The weights matter because a PINN is solving a multi-objective optimization problem: it must satisfy data anchoring, PDE residuals, symmetry constraints, and structural MHD constraints simultaneously.

7 Why the PINN benchmark remains scientifically honest

7.1 What is being enforced

The first PINN is enforcing:

- the exact initial state,
- periodicity,
- the reduced Alfvén dynamics,
- y-uniformity,
- background-state invariants,
- magnetic divergence control.

7.2 What is *not* yet being claimed

The first PINN is *not* yet claiming to solve:

- the full nonlinear two-dimensional ideal-MHD system,
- oblique wave interactions,
- compressive magnetosonic dynamics,
- shocklike structures,
- arbitrary boundary-value problems.

This honesty matters. A reduced benchmark PINN is still valuable, but only if it is described accurately.

8 Why the Alfvén benchmark is the right first PINN target

The benchmark has several strengths:

1. the reduced PDEs can be derived transparently from ideal MHD,
2. the exact solution is known,
3. the physically active fields are easy to interpret,
4. the benchmark is simple enough that training failure is informative rather than mysterious,
5. success provides a meaningful stepping stone toward richer MHD PINN formulations.

Bottom line. The first PINN in this project is not meant to be the final word on MHD neural solvers. It is meant to be the first controlled experiment in which a neural network is asked to satisfy a clean, derivable, and physically interpretable MHD subsystem.

9 What Module 3 accomplishes

Module 3 converts the Alfvén benchmark from a finite-volume verification problem into a PINN problem. It does so by:

- deriving the reduced benchmark residuals from ideal MHD,
- defining exactly which fields the network predicts,
- constructing loss terms that encode the benchmark physics and symmetry,
- clarifying what the first PINN stage does and does not claim.

That makes the PINN stage interpretable, technically manageable, and consistent with the broader long-term architecture of the project.